

A Multi-Scale Path Planning Optimization Framework for Planetary Rovers in Uncertain Terrain

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Planetary rover navigation must account for uncertain terrain, elevation-driven energy costs, and limited computational and power resources. Traditional shortest-path algorithms alone are not sufficient in this setting; they cannot capture the complexity of visiting multiple sites, handling complex constraints, or adapting to uncertainty in local terrain conditions. In this work, we develop a three-layer hierarchical planning framework that integrates global-level optimization with uncertainty-aware local planning. At the global level, we formulate a Mixed-Integer Linear Program (MILP) to optimize route selection under elevation-based movement costs estimated by satellite images, while enforcing feasibility through subtour elimination constraints. To bridge the discrepancy between low-resolution orbital data and fine-scale local terrain variability, we introduce an adaptive local planning layer modeled as a Markov Decision Process (MDP) on a refined grid, where elevation values are obtained through interpolation and associated with stochastic uncertainty. The motivation for this approach lies in the critical importance of energy efficiency and reliability in planetary missions, where even small improvements in routing can extend mission lifetime and increase scientific return. The results indicate that the hierarchical structure enables substantial reductions in computational complexity compared to repeated global MILP replanning approaches, making the method tractable for realistic missions.

I. Introduction

THE problem of planetary rover routing presents a unique set of challenges marked by unpredictability in terrain, the uncertainty in low-resolution maps, limited on-board computation, and the stringent demands of space exploration campaigns. Unlike terrestrial robotics, planetary rovers must also avoid large variations in elevation and incomplete knowledge of the environment, suggesting the need for optimization frameworks that explicitly model energy costs, uncertainty, and multiple sites of interest rather than relying solely on traditional shortest-path methods. Additionally, traversal on planetary surfaces is driven not only by geometric distance but also by elevation-induced energy costs and the finite energy and computational resources of the rover; small differences in route energy can affect mission lifetime significantly. This makes the development of optimization-based planning frameworks essential for enabling safe, efficient, and autonomous rover operations on extraterrestrial surfaces.

In the broader context of space exploration, optimization plays a central role in mission success. This is evident in operational systems such as SpaceX’s use of customized onboard convex optimization software for guidance [1] and Mars 2020 onboard scheduling [2], and optimization of space logistics and mission planning has been an active research field [3–9]. Since rovers operate under severe resource limitations (fuel, energy, time, etc.), optimization is essential to have a feasible and successful mission. Without optimization, space missions risk inefficiencies that can shorten mission duration, reduce scientific return, or compromise safety. For rover routing in particular, effective planning strategies must account for both global and local routing to ensure overall mission efficiency and safe navigation under these uncertainties.

Global path planning, assumed to be computed *before* the rover operations, focuses on optimizing routes (i.e., the optimal order of the visited sites) over large distances and provides overall coherence in the mission. This problem can often be formulated as a Mixed-Integer Linear Program (MILP) in a similar manner to the Traveling Salesman

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Problem (TSP). However, low-resolution maps from satellite terrain data alone cannot capture uncertainties that the rover might face *during* its operations, such as small obstacles, slight fluctuations in elevation, partial observability, and other hazards in the terrain, which can have a significant impact on the mission [10]. In contrast, local path planning emphasizes short-term decision-making in the presence of incomplete and changing information and often relies on immediate rover sensor data, which can also have inherent measurement errors [11]. This reflects a distinction between combinatorial route optimization at the mission level and shortest-path computation at the local level.

Existing approaches typically address only subsets of these challenges. Deterministic methods, such as MILP, allow for the computation of globally optimal paths that minimize the total terrain-based energy costs, enforce traversal constraints, and prevent infeasible routes. While MILP formulations provide a rigorous framework for global planning, their reliance on deterministic terrain data can limit performance when facing uncertain or partially observed landscapes. Existing local path planning algorithms, including classical graph-based methods such as Dijkstra and A* [12], are computationally efficient for identifying shortest paths given a well-defined cost structure and are suitable for near-term decision-making [13]. However, both Dijkstra and A* are limited to single-source, single-destination routing. Although A* efficiently computes optimal paths between a given start and end goal, it cannot determine the optimal sequence in which multiple target sites should be visited. The rover routing problem considered here requires selecting and ordering over a set of target locations, which introduces a combinatorial decision space over permutations of visits. Even when the route does not require returning to the starting location, this remains a variant of the TSP. Therefore, A* alone is insufficient for global mission planning and is instead used here to compute pairwise costs, while a MILP formulation is used to optimize the visitation sequence. That is, although they can effectively compute optimal sub-paths for a chosen pair of target sites, they lack the ability to handle complex combinatorial constraints at the global level, such as determining optimal visitation sequences over multiple scientific targets [14, 15].

More advanced stochastic planning frameworks, including Partially Observable Markov Decision Processes (POMDPs) and Model Predictive Control (MPC) have been explored in robotics to address uncertainty and dynamic-decision-making. While these methods with mild assumptions may provide guarantees for optimal decision-making under uncertainty, they are often computationally intractable for large-scale problems such as planetary exploration, particularly when long planning horizons or large state spaces are involved [16, 17]. Furthermore, many studies that involve MDP and MPC for route planning simply consider the local route planning scenario and do not perform global route optimization [18, 19]. Recent work has also explored the use of machine learning (ML) to address uncertainty in planetary navigation. For example, probabilistic learning-based approaches model terrain traversability by predicting distributions over traversal costs, such as slip, directly from terrain features [20]. Although this approach can improve robustness to unseen terrain, it is based on the accuracy and generalization of the learned models, which can fail in environments that differ from training data. While ML path planning algorithms can effectively deal with complex dynamic environments, they require complex training processes and large amounts of data [13]. In contrast, this work adopts a model-based optimization framework that does not rely on prior learning of terrain uncertainty or large amounts of training data. Instead, uncertainty is revealed in the local path planning stage. This allows the framework to remain robust to previously unseen terrain features without requiring pre-trained models or strict assumptions about the underlying distribution of data.

The current work addresses this by implementing a three-layer hierarchical planning framework. In the first layer, A* is applied on a coarse elevation grid to compute pairwise minimum-energy costs between all target nodes, explicitly incorporating elevation-based gravitational movement costs. These costs feed into a second layer consisting of a MILP formulation of the open TSP, which determines the globally optimal site visitation sequence using modified Miller-Tucker-Zemlin subtour elimination constraints. A third layer then refines each tour segment locally using an MDP framework defined on an adaptive bilinear-interpolation grid, accommodating uncertainty in terrain elevation through a stochastic interpolation penalty and enabling adaptive local decision-making [21]. This hierarchical decomposition preserves the computational efficiency of A* for pairwise routing while achieving optimality with respect to the deterministic MILP formulation of the mission sequencing problem. This coupling is essential, since the MILP provides an optimal visitation sequence with respect to the deterministic cost estimates derived from the coarse grid, while the MDP layer helps ensure that each segment of this sequence remains feasible and flexible when executed under fine-scale terrain uncertainty.

The computational results illustrate that the proposed hierarchical framework improves computational tractability by decomposing the problem into global and local components, thereby avoiding the need to solve a large-scale integrated stochastic optimization problem. The adaptive local planning layer enables the rover to avoid obstacles and high-cost terrain features that are not captured in coarse-resolution data, leading to reductions in traversal cost compared to fixed-path execution. At the same time, the hierarchical structure avoids the need to repeatedly solve large-scale

stochastic MILPs, resulting in a scalable and practical framework. This work situates itself at the intersection of operations research, autonomous navigation, and planetary robotics. By integrating deterministic MILP formulations with stochastic local planning strategies, this work advances the discourse on energy-efficient and resilient path planning for rovers operating on extraterrestrial surfaces.

The remainder of this paper is structured as follows. In Section 2, the proposed three-layer framework is presented, including the optimization problem formulation and the MDP-based local path planning. Section 3 presents and analyzes the results, including comparisons with deterministic baselines. Finally, Section 4 concludes the paper and discusses future directions for this work.

II. Problem Definition and Background

We present a three-layer hierarchical planning framework for autonomous surface navigation on terrain with uncertain elevation profiles. The framework integrates global route optimization with stochastic local path planning to address the uncertainty arising from imperfect terrain knowledge obtained from the satellite orbiter.

A. Problem Formulation

The operating environment is represented as a discrete elevation grid derived from orbital satellite data. Let X and Y denote the number of rows and columns of the coarse grid G^c consisting of nodes indexed by (i, j) where $i \in \{0, \dots, X - 1\}$ and $j \in \{0, \dots, Y - 1\}$ denote row and column indices, respectively, and $Z(i, j) \in \mathbb{R}$ denotes the elevation at each node. The traversal cost between adjacent nodes is modeled as the absolute change in gravitational potential: $c(s, s') = m \cdot g \cdot |Z(s') - Z(s)|$ where m is the vehicle mass, g is the gravitational acceleration, and s, s' are adjacent grid states. To model impassable terrain features such as boulders or craters in the local path planning scenario, a random subset of grid nodes is assigned prohibitively large elevation values ($Z = 10^6$), effectively acting as obstacles during path search. We also consider a set of scientific target sites that the rover must visit in its path.

Several constraints relevant to planetary rover routing can be adapted from the well-studied TSP [22]. In its classical form, the TSP requires visiting a set of nodes exactly once while minimizing the total traversal cost. However, the rover routing problem on planetary surfaces introduces unique challenges: not all nodes need to be visited, the sequence of visitation must often be optimized over a subset of scientifically valuable sites, and the traversal costs depend on energy-intensive elevation profiles rather than simple Euclidean distances.

To navigate planetary surfaces with accuracy, path planning cannot be considered in purely global or local terms. Global planning ensures mission-level coherence, determining the order and connections between sites, while local planning enables adaptation to real-time features [23]. As a result, a hybrid framework is essential. The global plan provides strategic overall guidance, while local adaptations are necessary to adapt in stochastic terrain conditions.

B. Layer 1: Global Path Planning with A* Search

The first layer computes the shortest paths between all pairs of mission target sites on the coarse grid G^c . Given a set of target sites, the A* algorithm is applied to each pair of sites to populate a pairwise cost matrix C and a corresponding path matrix P . The A* search employs an 8-direction connected neighborhood (allowing diagonal movement) and a Manhattan-distance heuristic scaled by the minimum observed traversal cost. The heuristic is defined as $h(s, g) = \alpha d(s, g)$, where $d(s, g)$ is the Manhattan distance and α is the minimum elevation cost. The resulting cost and path matrices serve as inputs to the global tour optimizer in Layer 2.

C. Layer 2: Global Tour Optimization via Mixed-Integer Linear Programming

The second layer solves a TSP over the set of scientific target sites using the pairwise cost matrix C produced by layer 1. Unlike shortest-path problems, this formulation determines the optimal ordering of target visits rather than the path between a fixed start and end node. The problem is formulated as a MILP and solved with the Gurobi optimizer with Pyomo. It differs from the classical TSP because the rover does not need to come back to the starting point, and the terminal node is not fixed but determined by the optimizer.

MILP provides an optimization framework for dealing with this situation. The objective function minimizes cumulative energy consumption, where arc costs between nodes are derived from gravitational work calculations based on terrain elevation differences. The constraints enforce path feasibility and subtour elimination. A modified Miller-Tucker-Zemlin formulation [24] is adopted for the subtour elimination constraints to prevent disconnected cycles in the rover's path. While MILP formulations may be computationally demanding for these large problems, their ability

to capture these mission-specific requirements makes them well suited for strategic, high-level rover planning. The following was formulated for a deterministic MILP:

Sets and Parameters

- Let $N = \{1, 2, \dots, n\}$ be the set of scientific target sites.
- Let c_{ij} denote the precomputed cost of traveling from site i to site j , obtained from the coarse-grid A* shortest path (Layer 1).
- Let $s \in N$ denote the designated start node.

Decision Variables

$$x_{ij} = \begin{cases} 1 & \text{if the rover travels directly from node } i \text{ to node } j \\ 0 & \text{otherwise} \end{cases} \quad \forall i \neq j$$

$$u_i \in \mathbb{Z}, \quad 0 \leq u_i \leq n, \quad \forall i \in N$$

where u_i is the ordering variable used for subtour elimination (MTZ formulation).

Objective Function

$$\min \sum_{i \in N} \sum_{\substack{j \in N \\ j \neq i}} c_{ij} x_{ij}$$

Constraints

1. Leave constraints The rover must leave the start node exactly once and may leave any other node at most once:

$$\sum_{\substack{j \in N \\ j \neq i}} x_{ij} = \begin{cases} 1 & i = s \\ \leq 1 & i \neq s \end{cases}$$

2. Enter constraints All nodes except the start must be entered exactly once, and the start node is not entered:

$$\sum_{\substack{i \in N \\ i \neq j}} x_{ij} = \begin{cases} 0 & j = s \\ 1 & j \neq s \end{cases}$$

3. Flow consistency constraint Ensures path connectivity when paired with subtour elimination:

$$\sum_{\substack{j \in N \\ j \neq k}} x_{kj} \leq \sum_{\substack{i \in N \\ i \neq k}} x_{ik}, \quad \forall k \in N, k \neq s$$

4. Subtour Elimination (MTZ)

$$u_j \geq u_i + 1 - n(1 - x_{ij}), \quad \forall i \neq j, i, j \in N, j \neq s, i \neq s$$

$$u_s = 0$$

5. Binary constraint

$$x_{ij} \in \{0, 1\}, \quad \forall i, j \in N, i \neq j$$

D. Layer 3: Stochastic Local Planning on an Adaptive Grid with MDP

The stochastic local path planning MDP is necessary to account for the uncertainty in both satellite data and the terrain. Orbital data provide only coarse prior estimates, and critical features over the planetary terrain are only observed locally through rover sensors. As such, this requires planning strategies that can accommodate uncertainty and update dynamically with new information. An adaptive grid is constructed on-demand for each segment. Only coarse cells traversed by or adjacent to the Layer 1 A* path are refined, limiting memory usage. Fine grid nodes attain elevation values from bilinear interpolation of their enclosing coarse cell.

The third layer refines each inter-waypoint segment of the global tour by performing local path planning on a spatially refined grid. This layer explicitly accounts for the uncertainty introduced by bilinear interpolation of elevation values at sub-coarse-grid resolution.

1. Adaptive Grid Construction and Bilinear Interpolation

For each tour segment, the relevant portion of G^c is expanded into a fine grid G^f by applying a uniform refinement factor $r \in \mathbb{Z}$ to each coarse cell. We have that $|G^f| = [(X-1)r+1] \cdot [(Y-1)r+1]$. Elevation values at fine-grid nodes are computed with bilinear interpolation. We let the four corners of the containing coarse cell be $Z_{00}, Z_{10}, Z_{01}, Z_{11}$. The interpolated elevation is $Z'(x, y) = (1-\alpha)(1-\beta)Z_{00} + \alpha(1-\beta)Z_{10} + (1-\alpha)\beta Z_{01} + \alpha\beta Z_{11}$ where $\alpha, \beta \in [0, 1]$ represent the normalized position within the cell [25].

2. Stochastic Cost Modeling

The local path and interpolation introduce uncertainty in the terrain. To capture this, an interpolation penalty $\phi(s, s')$ is defined for transitions involving fine-grid nodes. The penalty is modeled as a Gaussian random variable: $\phi(s, s') \sim N(\mu_\phi, \sigma_\phi^2)$. With this added penalty, the expected transition cost now becomes

$$E[c(s, s')] = mg|Z'(s') - Z'(s)| + E[\phi(s, s')] \quad \text{where} \quad \phi(s, s') = \begin{cases} 0, & \text{coarse transitions} \\ \mu_\phi, & \text{fine transitions} \end{cases}$$

3. MDP Formulation

Local path planning on the adaptive grid is formulated as a Markov Decision Process: $M = (S, A, P, C)$ where

- S : grid states
- A : 8-directional (N, NE, E, SE, ...) movement actions
- P : deterministic transition model
- C : expected movement cost

Let $f(s, a)$ denote the deterministic transition function that maps a state-action pair to the resulting adjacent grid state under the chosen movement direction. The transitions satisfy

$$P(s'|s, a) = \begin{cases} 1, & \text{if } s' = f(s, a) \\ 0, & \text{otherwise} \end{cases}$$

Uncertainty is modeled exclusively in the cost function rather than the transition dynamics, resulting in a stochastic shortest-path problem with deterministic state evolution.

4. Local Path Planning with A*

Rather than solving the full MDP with dynamic programming (which is computationally prohibitive at high resolutions), paths between consecutive global targets are computed using A* search with the expected cost function. In particular, the A* search is applied to minimize $\sum E[c(s, s')]$. For each segment, the algorithm computes a refined path on the adaptive grid.

5. Deterministic Global Path in Stochastic Scenario

To allow a consistent comparison with the local planning method, each segment of the deterministic global coarse-grid path is projected onto the refined grid. The coarse plan solves the TSP over target sites, and each coarse

step covers multiple fine-grid nodes. To evaluate under stochastic terrain (interpolated fine nodes), every coarse step is expanded into a sequence of fine nodes.

Specifically, each coarse node is mapped to its corresponding location on the fine grid, and intermediate fine-grid states are inserted along each segment to produce a continuous trajectory at higher resolution. This is the exact sequence of nodes the rover would traverse if the coarse solution was executed at high resolution without any adaptation. The deterministic baseline corresponds to executing the interpolated path without the local path planning MDP. Although the path is fixed, its cost is evaluated under the stochastic model described above. Importantly, this baseline does not adapt to fine-scale terrain variations, newly introduced obstacles, stochastic realization of costs.

Let the coarse path be defined as $P^C = \{(i_0, j_0), \dots, (i_k, j_k)\}$. Given a refinement factor r , each coarse grid cell is refined such that a coarse node (i, j) corresponds to the fine-grid node: $(i^f, j^f) = (ri, rj)$. Then, for each consecutive pair of nodes, a fine-grid path is constructed by incrementally stepping between their corresponding fine-grid coordinates resulting in a refined path. This preserves the structure of the original path while increasing its resolution.

III. Results

A. Optimal Path in the Deterministic Coarse Grid Case

The following figures are a result of using a grid size of 178×30 with a total of 5340 coarse grid nodes. The model was applied on arbitrary elevation data following the format of (x, y, z) obtained from a data extraction tool made by the Scripps Institution of Oceanography [26]. In addition, 12 target sites are used as locations the rover must visit in its path. All numerical experiments were conducted on a MacBook Pro equipped with an Apple M4 processor running macOS (Darwin 25.3.0). The proposed framework was implemented in Python (CPython 3.12.2), with optimization models formulated using the Pyomo library (version 6.9.2) [27, 28]. MILP problems were solved using the Gurobi Optimizer (version 11.0.0) [29]. Default solver parameters were used, and the MILP models were solved to optimality subject to standard feasibility and optimality tolerances.

Figures 1 and 2 show the optimal path resulting from the execution of Layer 1 (A*) and Layer 2 (MILP). The rover attempts to find the most energy efficient path, avoiding elevation while also meeting the constraint of visiting the target sites. For this case, 12 target sites are considered, and the time it took for layer 1 to complete was 2.94 seconds, and it took Layer 2 (the TSP MILP) 0.44s to determine the optimal path. The total deterministic coarse tour cost from this scenario is 85690.35.

B. Stochastic Local Planning

Next, we consider the stochastic local path planning with bilinear interpolation (Layer 3) and the addition of obstacles. For this scenario, a refinement factor of $r = 3$ (each coarse cell becomes 3×3 fine cells) was chosen and 20 obstacles were placed. The obstacles are explicitly incorporated to reflect the realities of local path planning, where the rover must dynamically respond to unforeseen hazards in its immediate vicinity (e.g. a large boulder or rocks). These obstacles are strategically placed along and around the nominal deterministic path to evaluate the effectiveness of the local planning algorithm, specifically its ability to detect, avoid, and re-route around obstructions in real time.

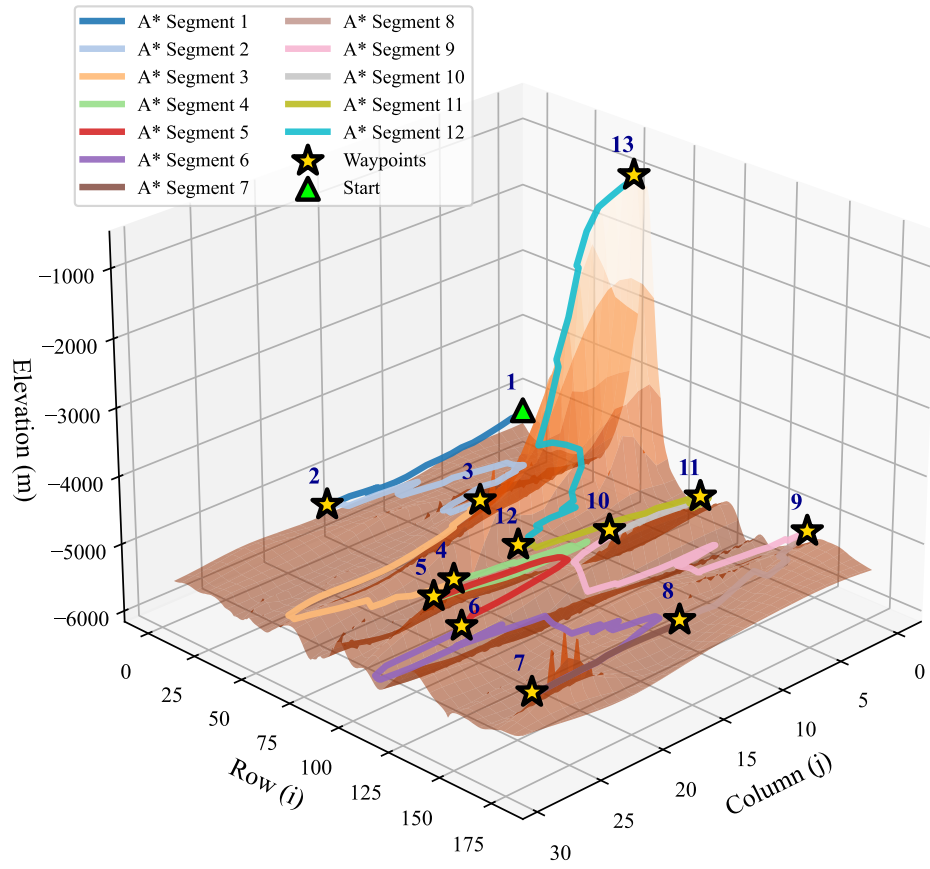


Fig. 1 Deterministic Optimal Paths (Three-Dimensional)

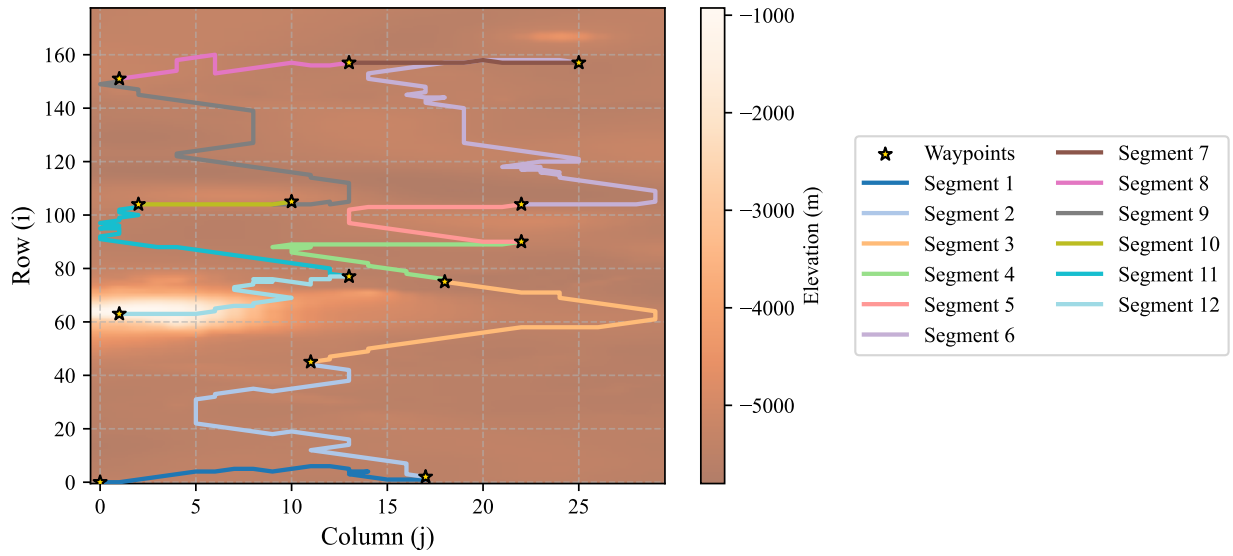


Fig. 2 Deterministic Optimal Paths (Two-Dimensional)

Table 1 Segment-wise comparison of coarse and refined path costs. Δ denotes the difference between expected refined cost and coarse cost.

Seg	Fine Path Length	Coarse Cost	Elev Cost	Penalty	Δ (%)	Time (s)
1	54	1373.40	1430.08	26.50	+83.18 (6.06)	0.076
2	140	3217.68	3357.20	69.50	+209.02 (6.50)	0.282
3	111	4483.17	4478.81	55.00	+50.64 (1.13)	0.323
4	77	3227.49	3114.13	38.00	-75.36 (-2.33)	0.161
5	70	3610.08	3631.88	34.50	+56.30 (1.56)	0.153
6	185	7416.36	7239.78	92.00	-84.58 (-1.14)	0.510
7	37	1118.34	1079.10	18.00	-21.24 (-1.90)	0.035
8	37	2589.84	2589.84	18.00	+18.00 (0.70)	0.227
9	139	9947.34	9794.74	69.00	-83.60 (-0.84)	0.471
10	25	1451.88	1451.88	12.00	+12.00 (0.83)	0.011
11	90	3737.61	3774.67	44.50	+81.56 (2.18)	0.133
12	52	43517.16	43517.16	25.50	+25.50 (0.06)	1.016

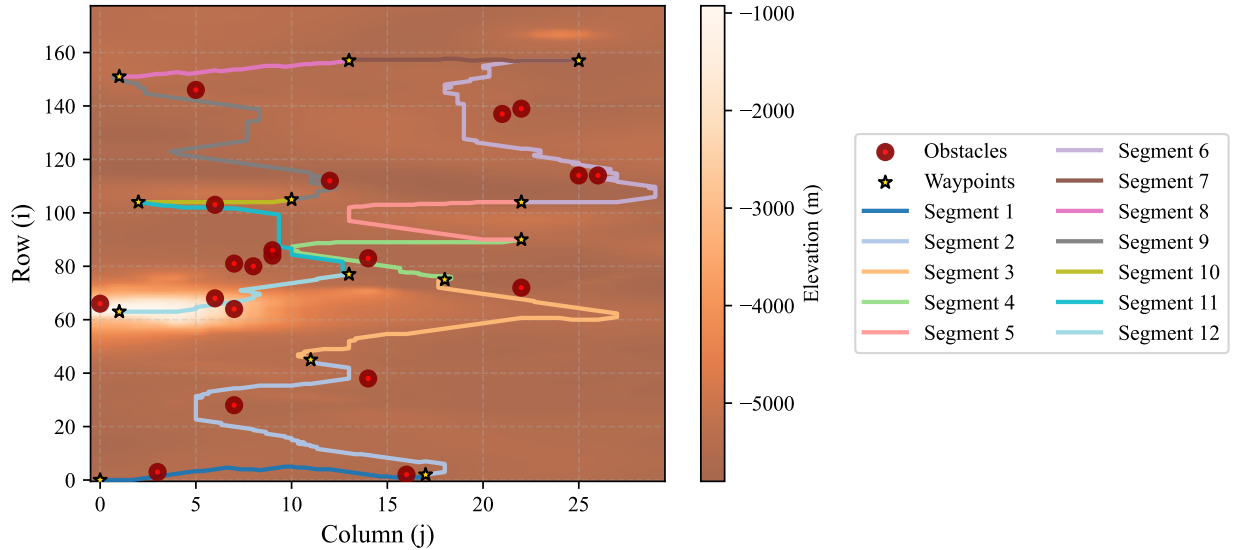


Fig. 3 Stochastic Local Path Planning Paths

As seen in Fig. 3, the rover weaves around the obstacles it encounters in its local path while also visiting the target sites. Table 1 above shows how the refined planner consistently adjusts the deterministic baseline by incorporating local terrain detail and uncertainty, typically increasing cost due to risk penalization while occasionally reducing cost through improved local routing. Importantly, the refined local planner provides a more realistic and risk-aware estimate of traversal cost, correcting optimistic biases in the coarse grid scenario while occasionally improving suboptimal segments through its higher-resolution adaptation.

Additionally, the local path planning required 3.40 seconds, resulting in a total runtime of 6.77 seconds across all three layers. This stage naturally incurs higher computational cost due to the increased resolution of the fine grid and incorporation of obstacles. However, the overall runtime remains modest, especially when compared to an alternative approach that repeatedly resolves a global MILP as new information is realized. Rather than a formal stochastic MILP, this corresponds to repeatedly solving a large-scale deterministic MILP as uncertainty is incrementally revealed, leading to a substantial increase in problem size and computational complexity. The hierarchical framework resulted in a model with 182 variables and constraints, whereas solving a global MILP at each re-planning step resulted in a model with 52,155 variables and 62,827 constraints. These results demonstrate that the proposed hierarchical framework enables adaptive high-resolution planning while maintaining computational efficiency.

Table 2 Cost Comparison of fixed path and adaptive local planning method under the stochastic cost model

Method	Cost	Obstacle Penalty
Deterministic (No Re-optimization)	174663.30	+30689657.41
Adaptive (Local Replanning)	171421.04	+ 0.00

C. Deterministic Baseline Comparison with Adaptive Local Planning

We now compare the total tour cost of the deterministic baseline and the adaptive local planning method under the stochastic cost model. Table 2 shows that the adaptive method achieves a reduction in total cost under the stochastic cost model, improving over the baseline. These results suggest that global plans can become invalid or too expensive when local conditions change.

The large performance gap is primarily driven by the baseline’s inability to adapt to refined terrain conditions. Because the deterministic path is fixed after global planning, it may intersect obstacles or high-gradient regions introduced at the fine scale, which results in extremely large elevation costs. In contrast, the adaptive method recomputes paths using updated terrain information with knowledge of obstacles and avoids such regions, yielding substantially lower expected costs. These results demonstrate that the deterministic baseline, while optimal under coarser assumptions, becomes highly suboptimal when executed in a more detailed and uncertain environment. In contrast, the local re-planning with MDP enables the rover to respond well to uncertainties. Overall, adaptive local planning significantly improves both performance and robustness.

IV. Conclusion

Planetary rover routing is a challenging problem due to uncertainty in terrain information and the high cost of navigation errors, such as encountering obstacles or steep gradients that are not known in advance. Traditional deterministic approaches assume complete knowledge of the environment, which is often unrealistic in planetary exploration settings. This motivates the need for methods that can efficiently incorporate uncertainty and adapt to new information during execution.

This work presents a three-layer hierarchical framework for multi-site planetary rover routing involving uncertainty. The architecture integrates global optimization, deterministic path planning, and adaptive local decision-making with a Markov Decision Process (MDP). Specifically, we use the A* algorithm to compute pairwise costs, a Mixed-Integer Linear Program (MILP) to determine the optimal visit order, and an MDP-based method for local path planning on a finer grid.

A key advantage of this framework is the reduction in computational complexity by avoiding the need to solve a large-scale integrated stochastic optimization problem. Rather than performing an expensive stochastic optimization and resolving a MILP every few steps, we use an MDP based method locally. The timing results show that each layer runs efficiently, and that the overall pipeline is practical. Additionally, the results show that deterministic planning can break down in uncertain environments. When the coarse path is followed without re-optimization under local uncertainty, the path can become infeasible or incur very large costs because it does not account for incomplete terrain information or newly introduced obstacles. In contrast, the MDP-based local planner adapts to these changes and finds feasible paths that avoid obstacles. This leads to a significant improvement in total cost, and additionally shows that adaptability is crucial in uncertain and partially observable environments.

Overall, this work shows that combining global optimization with local stochastic planning is an effective way to tackle the problem of planetary rover routing. The framework is computationally efficient and produces more flexible and realistic solutions than a purely deterministic approach.

There are several directions for future work. For example, learning-based approaches in conjunction with global optimization may be useful in getting better estimates of terrain uncertainty in real time. Additionally, extending the framework to incorporate more complex rover dynamics, energy and power depletion/recharging constraints, or multi-rover coordination could improve its applicability to real missions. Finally, integrating reinforcement learning or more advanced control methods may further enhance the performance of the local planner.

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